

2021 ELECTIVE MATHEMATICS PAPER 2

1.

The polynomial $f(x) = 2x^3 + px + qx - 5$ has $(x-1)$ as a factor and a remainder of 27 when divided by $(x + 2)$, where p and q are constants. Find the values of p and q .

2.

Evaluate: $9 \int_1^9 \frac{x(2x-3)}{\sqrt{x}} dx$

3.

Given that $(p + 1/2\sqrt{3})(1 - \sqrt{3})^2 = 3 - \sqrt{3}$,

find the value of p .

4.

${}^5P_2 = 190$, find the value of y

5.

The table shows the distribution of monthly income (in thousands of naira) of workers in a factory

Monthly Income (GHC1000)	135-139	140-149	150-154	155-164	165-169
Number of workers	20	42	28	38	22

(a) Draw a histogram for the distribution.

(b) Use your graph to estimate the mode of the distribution.

6.

A bag contains 24 mangoes out of which six are bad. If 6 mangoes are selected randomly from the bag with replacement, find the probability that not more than 3 are bad.

7.

(a) The speed of a moving bus reduced from 45m/s to 5m/s with a uniform retardation of 10m/s^2 . Calculate the distance covered.

(b) A bucket full of water with mass 16kg is pulled out of a well with a light inextensible rope. Find its acceleration when the tension in the rope is 240N. [Take $g = 10\text{m/s}^2$]

8.

Given that $x = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$ and $y = \begin{pmatrix} -9 \\ 15 \end{pmatrix}$ calculate, correct to the nearest degree, the angle between the vectors

9.

(a) A jogger is training for 15km charity race. He starts with a run of 500 metres, then he increases the distance he runs daily by 250 metres.

(i) How many days will it take the jogger to reach a distance of 15km in training?

(ii) Calculate the total distance he would have run in the training.

(b) The second term of a Geometric Progression (GP) is -3. If its sum to infinity is $25/2$, find its common ratios.

10.

P and Q are two linear transformations in the X-Y plane defined by

P: $(x, y) \rightarrow (-3x + 6y, 4x + y)$ and

Q: $(x, y) \rightarrow (2x - 3y, -4x - 6y)$.

(a) Write down the matrices of P and Q. (b) What is the image of $(-2, -3)$ under the transformation Q?

(c) Obtain a single transformation representing the transformation Q followed by P.

(d) Find the image of $(1, 4)$ when transformed by Q followed by P.

(e) Find the image P^1 of the point $(-\sqrt{2}, 2\sqrt{2})$ under an anticlockwise rotation of 225° about the origin.

11.

(a) Find the equation of the normal to the curve $y = (x^2 - x + 1)(x - 2)$ at the point where the curve cuts the X - axis.

(b) The coordinates of the points P, Q and R are $(-1, 2)$, $(5, 1)$ and $(3, -4)$ respectively. Find the equation of the line joining Q and the midpoint of line PR.

12.

A box contains 5 red, 7 blue and 4 green identical bulbs. Two bulbs are picked at random from the box without replacement.

Calculate the probability of picking: (a) same color of bulbs; (b) different color of bulbs (c) at least one red bulb.

13.

The table shows the frequency distribution of heights (in cm) of pupils in a certain school.

Heights	100-109	110-119	120-129	130-139	140-149	150-159	160-169
Frequency	27	58	130	105	50	25	5

(a) (i) Construct a cumulative frequency table.

(ii) Use the table to draw a cumulative frequency curve.

(b) Using the curve, estimate the:

(i) median height;

(ii) inter quartile range

(iii) percentage of students whose heights are most 130cm.

14.

The position vectors of P, Q and R with respect to the origin are $(4i-5j)$, $(i+3j)$ and $(-5i+2j)$ respectively. If PQRM is a parallelogram, find:

(a) the coordinates of M;

(b) the acute angle between P-M and P-Q, correct to the nearest degree.

15.

(a) A girl threw a stone horizontally with a velocity of 30m/s from the top of a cliff 50m high. How far from the foot of the cliff does the stone strike the ground? [Take $g = 10\text{m/s}^2$]

(b) A body A, of mass 2kg is held in equilibrium by means of two strings AP and AR. AP is inclined at 56° to the upward vertical and AR is horizontal.

Find the tensions T_1 , and T_2 , in the strings [Take $g = 10\text{ms}^2$]